Principles and Pitfalls in the Differential Diagnosis and Management of Childhood Obesities[1–3]

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ABSTRACT

Obesity is currently the most prevalent chronic childhood disease in Western countries. It is one of the most frequent consultations in general pediatrics and is even more common in pediatric endocrinology. As might be predicted, the prevalence of obesity-associated comorbidities is also increasing in children and adolescents. It is widely accepted that this increase in obesity results from an imbalance between energy intake and expenditure, with an increase in positive energy balance being closely associated with the current lifestyle in Western countries. However, there is increasing evidence indicating that an individual's genetic background is important in determining obesity risk. The physiologic mechanisms controlling appetite and energy expenditure are being revealed in part because of the identification of new causes of human monogenic, syndromic, and endocrine-related obesity. Thus, it is no longer appropriate to talk about obesity, but rather about "obesities" or "different diseases causing obesity," because their pathophysiologic bases differ. Moreover, these obesities require different diagnostic and management approaches. The pediatrician must be aware of this issue and focus the clinical history and physical examination toward specific clinical signs and symptoms to better exploit the available diagnostic and therapeutic resources when facing a child with obesity. Genetic, genomic, and metabolomic studies are often necessary to obtain a more appropriate diagnosis. Cognitive behavioral therapy is fundamental in obese children. The identification of potential targets will hopefully result in new pharmacologic approaches for translational and personalized medicine for obesity in the near future. *Adv. Nutr. 5: 299S–305S, 2014.*

Introduction

Childhood obesity has become one of the most serious concerns among pediatricians. The different pathologies that comprise what is referred to as "childhood obesity" are extraordinarily complex, and our knowledge is limited regarding their underlying causes. In this review we analyze the concept and epidemiologic, taxonomic, and clinical aspects of childhood obesity. We also discuss the differential diagnosis of the distinct types of obesities and the therapeutic approaches currently available.

Concept

Obesity is the excessive accumulation of adipose tissue that results in an increased risk of developing associated pathologies, including mortality. In children, a precise definition is difficult because the appearance of comorbidities can be delayed; however, BMI [weight (kg)/(height (m)^2], which shows a good correlation with body fat content (1,2), is commonly used. In children and adolescents, standardized values of BMI as a function of age and sex for their reference population must be used. This has generated intense controversy regarding the establishment of "cutoff points" and the reference populations that should be used (3,4). In general, a child has excess body fat when his/her BMI is greater than the 95th percentile for his/her age and sex (1). An optimal definition can be obtained by applying a cutoff point of +2 SDS above the estimated mean value for individuals of the same population, age, and sex, as proposed by the WHO (5). There is also no consensus regarding the definition of morbid obesity in children and adolescents. Some authors suggest a BMI ≥3 SDS or 200% of the ideal body weight for height as possible cutoff points.

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Epidemiology

The incidence of obesity has progressively increased during the past 30 y in all developed countries including the United States and most European countries, with these changes being more apparent in Mediterranean countries (with a prevalence of between 20% and 40%) than in northern countries. In Spain, the prevalence of obesity in children has increased, with recent cross-sectional studies reporting a joint prevalence of overweight and obesity of 27.6% in the population aged between 2 and 17 y (2,6,7).

These data stimulated diverse initiatives oriented toward prevention and early therapeutic intervention, including the elaboration of the "Clinical Practice Guide for the Prevention and Treatment of Childhood Obesity" (4). Research directed at understanding and treating this disease has also been reinforced, including the creation of the Centro de Investigación Biomédical en Red (CIBER) de la Fisiopatología de la Obesidad y Nutrición by the Institute of Health Carlos III. This is a stable network of investigators throughout Spain with the aim of studying obesity and nutrition from all aspects and in collaboration.

Classification

Obesity in children has diverse causes, which can result in radically different pathologic entities. Although it is accepted that the large increment in childhood obesity is the result of prolonged positive energy balance due to the Western lifestyle (polygenic obesity), some cases are due to genetics, endocrine diseases, or syndromes. Although more limited in number, physicians must be aware of the existence of these other causes.

Exogenous or "common" obesity. Exogenous or "common" obesity is the most frequent form of obesity in children and adolescents. Here hypercaloric nutrition and reduced physical activity coexist resulting in the accumulation of excess energy in the form of adipose tissue. Unfortunately, this is also influenced by the individual's socioeconomic situation. However, not all individuals exposed to the same "obesogenic" environment develop obesity or the same degree of obesity. This is due to the fact that these "exogenous" factors act upon an "endogenous" make-up, including the genetic information specific to each individual. This can explain, at least in part, the large heritable influences observed in obesity (8).

By studying large cohorts of individuals affected with a certain disease, genome-wide association studies aim to identify new genes, quantitative trait loci, or haplotypes that permit a better assessment of an individual's risk of developing a disease. In the latest "human obesity gene map," the number of genes and quantitative trait loci associated with an adiposity phenotype in murine models identified by genome-wide association studies reached 244 and 408, respectively (9). For example, variants in the primer intron of the gene coding for the fat mass and obesity–associated protein (FTO)[4] have been associated with an elevation in BMI equivalent to approximately +0.4 kg/m^2 for each risk allele (10). Epigenetic modifications, especially during early phases of development, also appear to play a relevant role in determining an individual's risk of developing obesity (11). Hence, this common form of obesity should be referred to as "polygenic obesity."

Thus, obesity in most affected children has a multifactorial etiology, with a polygenic basis. The polygenic basis per se has limited effect on the phenotype, and it is only in combination with other risk variables, especially environmental factors conducive to weight gain, that obesity develops. However, as mentioned above, a proportion of children are obese due to monogenic causes or associated with syndromes. These conditions should be diagnosed in the clinic because treatment methods can differ.

Monogenic obesity. Monogenic obesity results from an alteration in a single gene and represents a minority of the total population of obese children. A common feature of these patients is severe early-onset obesity. The forms of monogenic obesity known to date can be classified into the following 3 categories:

1. Genes involved in the adipocyte-hypothalamic system (leptin/melanocortin axis). The most important genes included in this category are those coding for leptin (*LEP*), leptin receptor (*LEPR*), proopiomelanocortin (*POMC*), proconvertase 1 (*PCSK1*), proconvertase 2 (*PCSK2*), melanocortin receptor 4 (*MC4R*), and peroxisome proliferator-activated receptor γ (*PPAR*γ). The number of patients reported to be obese due to alterations in these genes could represent ~7–8% of the different causes of obesity, with *MCR4* mutations being the most frequent and representing 2.5–3% of obese patients (3). The clinical features of patients affected by these types of monogenic obesities are summarized in **Table 1** (3,12).
2. Genes associated with hypothalamic development. Anomalies in at least 3 genes associated with hypothalamic development, including single-minded homolog 1 (*SIM1*), brain-derived neurotrophic factor (*BDNF*), and neurotrophic tyrosine receptor kinase 2 *(NTRK2*), have been shown to correspond to obesity (3). To date, the reported number of cases involving these genes is small.
3. Obesity associated with malformation syndromes. There are a number of syndromes in which obesity is among the characteristic feature (**Table 2**) (3,12).

Secondary obesity. Obesity can also be the consequence of a disease or its treatment. Endocrine pathologies and pathologic processes or therapeutic protocols that affect the pituitary-hypothalamic axis can promote weight gain. Likewise, some pharmacologic treatments, especially those with psychoactive ingredients, can result in obesity. The most frequent causes of secondary obesity and their most important characteristics are shown in Table 2.

Clinical Aspects of Childhood Obesity: Comorbidities

Excess adipose tissue can induce numerous alterations in different organs and systems; thus, the patient's history and physical

[4] Abbreviations used: *BDNF*, brain-derived neurotrophic factor; FTO, fat mass and obesity–associated protein; IGF-I, insulin-like growth factor I; IGFBP-3, insulin-like growth factor binding protein 3; IR, insulin resistance; *LEPR*, leptin receptor; *MC4R*, melanocortin receptor 4; MS, metabolic syndrome; *NTRK*, neurotrophic tyrosine receptor kinase; *PCSK1*, proprotein convertase type subtilisin/kexin 1; *POMC*, proopiomelanocortin; *PPAR*γ, peroxisome proliferator-activated receptor γ; *SIM1*, single-minded homolog 1.

TABLE 1 Clinical description of the main monogenic forms of obesity[1]

	Description
MC4R mutations	Severe early-onset obesity with hyperphagia; binge eating reported; tall stature and advanced bone age during childhood
Leptin deficiency	Severe early-onset obesity with low circulating leptin concentrations; hypogonadotrophic hypogonadism; reversible after recombinant leptin administration
Leptin receptor mutations	Severe early-onset obesity with hypogonadotrophic hypogonadism; low TRH and GnRH concentrations
POMC deficiency	Severe early-onset obesity; low plasma cortisol concentrations; common impairment of skin and hair pigmentation
PCSK1 deficiency	Severe early-onset obesity with increased *POMC* and proinsulin and low insulin and cortisol concentrations; hypogonadism possible

[1] GnRH, gonadotrophin-releasing hormone; *MC4R*, melanocortin receptor number 4; *PCSK1*, proprotein convertase type subtilisin/kexin 1; *POMC*, proopiomelanocortin; TRH, thyrotropin-releasing hormone.

examination and any necessary complementary investigations should be oriented toward the patient's signs and symptoms. The most important metabolic complication is insulin resistance (IR), which can be defined as the incapacity of circulating insulin, under normal conditions, to induce the uptake of peripheral glucose, suppress hepatic gluconeogenesis, and inhibit the production of VLDLs. This results in a compensatory increase in insulin secretion and can lead to carbohydrate intolerance and eventually type 2 diabetes when the compensatory capacity of the pancreas is no longer sufficient (13).

IR is considered the pathophysiologic basis of several alterations to carbohydrate and lipid metabolism and arterial hypertension, which tend to coexist in an individual and can lead to coronary artery disease. This is defined as metabolic syndrome (MS). After several changes in the criteria, the International Federation of Diabetes proposed a modification of the Adult Treatment Panel III criteria to include the requirement of a waist circumference greater than the reference value according to ethnic group, as a measurement of central obesity, in conjunction with 2 other criteria (14).

The diagnosis of MS in children is complicated because they are continuously developing and the absolute values used to establish pathologic limits of each variable in adults cannot be applied. Despite attempts to establish cutoff points according to percentile and adjusted for each age range, some authors doubt the reliability of this diagnosis and the benefit of applying this concept in children and support the identification and treatment of risk factors on an individual basis (15). The International Federation of Diabetes recommends that this diagnosis not be made in children <10 y of age (16).

Diagnosis

The measurement of total body fat and its distribution is important because different adipose depots have distinct roles in the development of metabolic complications. However, access to techniques for directly measuring body fat content is limited in normal clinical practice. Indirect estimations of visceral adipose content can be made by performing waist and hip measurements and by using international references that are classified according to ethnic group (17). Computerized axial tomography or MRI can also be used to directly and precisely quantify visceral fat.

Although most cases of childhood obesity are polygenic and secondary to a positive energy balance, after the presence of obesity has been established the physician should determine whether there are clinical features common to monogenic mutations (especially *MC4R*), endocrine diseases, or syndromes or specific treatments such as hormonal replacement of deficiencies (**Tables 1** and **2**). The family history should include all personal and familial information that could help to determine the etiology of the patient's obesity, with special emphasis placed on nutritional habits and physical activity of both the child and his or her family (18). A general pediatric examination should be performed but

TABLE 2 Genetic syndromes and disease states commonly presenting with obesity[1]

Genetic syndrome	Secondary obesity
Alstrom-Wolfram (OMIM: 203800)	Hypothyroidism
Bardet-Biedl (OMIM: 209900)	Hypercortisolism
Beckwith-Wiedemann (OMIM: 130650)	Pseudohypoparathyroidism (Albright's hereditary osteodystrophy)
Börjesson-Forssman-Lehmann (OMIM: 301900)	Neonatal hyperinsulinism
Carpenter (OMIM: 201000)	Growth hormone deficiency
Cohen (OMIM: 216550)	Hypothalamic obesity
Down (OMIM: 190685)	Iatrogenic obesity
MEHMO (OMIM: 300148)	
MOMO (OMIM:157980)	
Prader-Willi (OMIM:176270)	
Smith-Magenis (OMIM: 182290)	
Wilson-Turner (OMIM: 309585)	
WAGRO (OMIM: 612469)	

[1] OMIM, On-line Mendelian Inheritance in Man database (http://www.ncbi.nlm.nih.gov/sites/entrez?db = omim). MEHMO, mental retardation, epileptic seizures, hypogonadism and hypogenitalism, microcephaly, and obesity; MOMO, macrosomia, obesity, macrocephaly, and ocular abnormalities; WAGRO, Wilms tumor, anirida, genitourinary anomalies, mental regardation, and obesity syndrome.

specifically directed toward the detection of whatever sign that could indicate the cause of the obesity and the existence of associated comorbidities.

Complementary Studies

Investigations in obese children should include measurement of circulating concentrations of transaminases (aspartate transaminase and alanine transaminase), basal glycemia, and a lipid profile (19). Because the appearance of IR and subsequent alterations in carbohydrate and lipid metabolism is gradual (20), it is possible that hyperinsulinemia may exist in the absence of alterations in fasting glycemia (21). Thus, it is advisable to analyze basal insulinemia in obese children and adolescents and to calculate the HOMA index [fasting glucose (mmol/L) × insulin (μU/mL)/22.5] as an indicator of IR.

Whether other complementary studies should be performed is determined by the relevant data obtained in the clinical history and physical examination (Fig. 1), including the following:

- Radiography of the left hand and wrist. Acceleration of bone age in respect to chronologic age, but adequate for the child's height, is frequent in obesity.
- Biochemical investigations. Evaluation of electrolytes because sodium and potassium concentrations can be altered in hypercortisolism, calcium and phosphorus in pseudohypoparathyroidism, and urea and creatinine with the presence of microalbuminuria in nephropathies. It is also recommended that a hematogram and iron and ferritin concentrations be studied because of possible associated deficiencies (22).
- Oral-glucose-tolerance test. When the patient belongs to an ethnic group with an elevated risk (Hispanic, African American) or there are alterations in fasting glycemia (>100 mg/dL) or insulinemia (>15 μU/mL), dyslipidemia, hypertension, family history of type 2 diabetes, or conditions associated with IR such as *Acanthosis nigricans* or symptoms of polycystic ovary syndrome, an oral-glucose-tolerance test should be performed (23).

Hormonal studies should be performed to rule out the possibility of hypothyroidism [free thyroxin (T4) and thyroid stimulating hormone] or hypercortisolism (24-h urinary cortisol concentrations). If growth hormone deficiency or insensitivity is suspected, serum insulin-like growth factor I (IGF-I) and insulin-like growth factor binding protein-3 (IGFBP-3) concentrations should be determined. Circulating concentrations of adiponectin, in particular its high-molecular-weight multimer, have been proposed as a possible indicator of the risk of developing alterations in carbohydrate metabolism because of its inverse relation with IR (24).

Abdominal ultrasound is the technique of choice if nonalcoholic steatohepatitis or polycystic ovary syndrome is suspected (25). Signs or symptoms that suggest specific comorbidities will determine whether additional tests, such as a psychological evaluation or further specialized tests (digestive, cardiac, orthopedic, nephrologic, etc.), or other complementary examinations are needed.

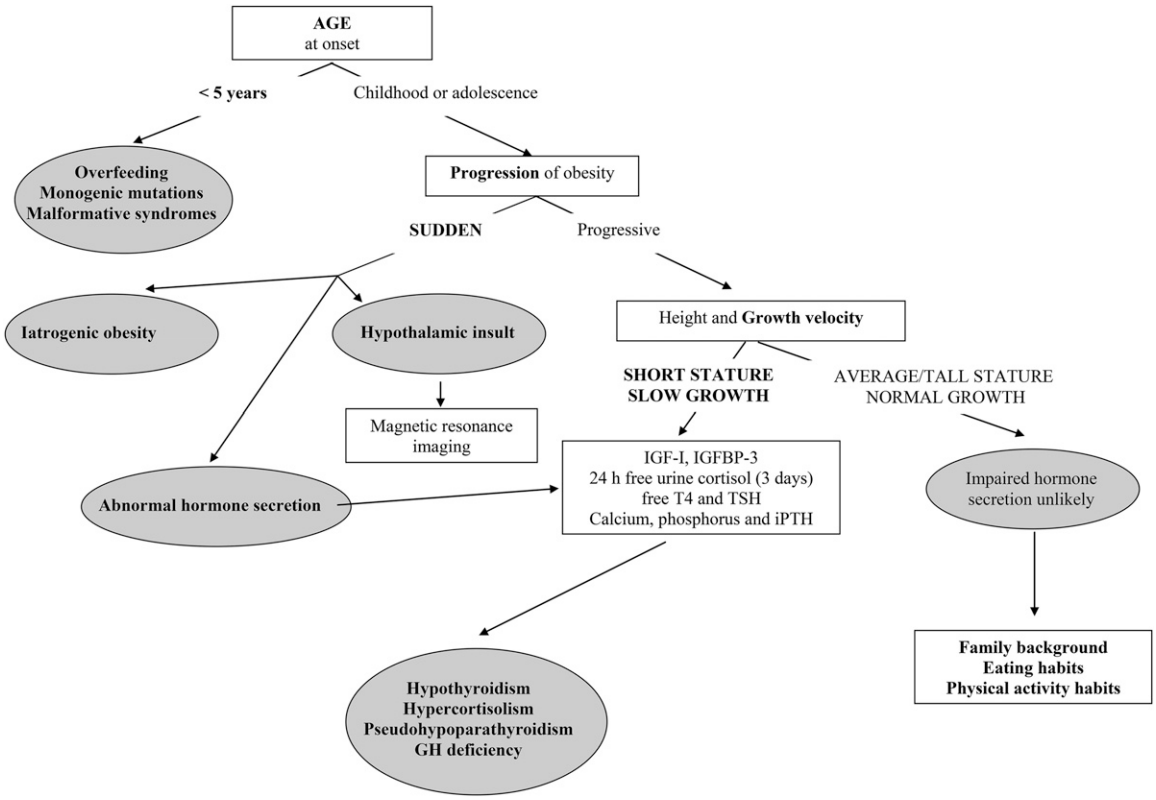

FIGURE 1 Clinical algorithm for the study of childhood obesity. The different steps to be followed for the differential diagnosis of obesity in children are shown. GH, growth hormone; IGF-I, insulin-like growth factor; IGFBP-3, insulin-like growth factor binding protein 3; iPTH, intact parathyroid hormone; T4, thyroxin; TSH, thyroid stimulating hormone.

Therapeutic Strategies in Childhood Obesity

The treatment of obesity in children is based on the reorganization of dietary habits, increased physical activity, and behavioral modification. In this age group, the indications for pharmacologic or surgical treatment remain limited, and it is impossible to establish specific recommendations for their use or to indicate their long-term results (26).

Treatment of the obese child should be supervised by an experienced professional with frequent visits, especially during the initial treatment period. An extended follow-up period is required and can include individualized programs and/or group therapy, depending on the availability of professionals and installations. Sufficient data are not available at present to make recommendations regarding group or individualized therapy or the timing, frequency, or length of visits. It is recommended that treatment be individualized on the basis of the characteristics of the child and his or her family.

Behavioral treatment

The objective of this treatment is to help the child acquire new abilities that will allow him or her to reach previously agreed-upon goals. It consists of 2 fundamental components: techniques to modify behavior and therapy to reduce stress.

Techniques for behavioral modification. Classical or respondent conditioning is used, with food intake being the conditioned response and the stimulus being situations that the patient associates with increased food intake (27).

Stress management therapy. This cognitive approach allows objectives to be established and realistic evaluation of successful changes in diet and behavior (28). Self-monitoring is of utmost importance, in addition to the control of stimuli, cognitive restructuring, solving of problems that arise during therapy, and the prevention of relapses (28). Adequate and sufficient interaction with the pediatrician results in what is called "motivational interview," which is oriented at achieving set goals, with satisfactory results achieved especially when the family is implicated (29).

Clinical practice guidelines indicate that, for weight reduction, sustained behavioral changes must be achieved (degree of recommendation: D) as part of a combined intervention (degree of recommendation: B) using family-targeted strategies (degree of recommendation: B), with specific intervention needed according to the age range (expert opinion) (4,30).

Nutritional intervention. Nutritional intervention includes combined strategies aimed at achieving changes in lifestyle. It is impossible to establish specific recommendations regarding nutritional intervention in childhood obesity, because controlled long-term follow-up are still required, including evaluation of diets with different degrees of caloric restriction and alterations in macronutrient composition (limitation of fats and carbohydrates or increases in fiber or proteins) or micronutrients (calcium). Diets with alterations in the proportions of macronutrients are of interest because low-carbohydrate diets with a low glycemic index have been shown to be as effective for short-term weight loss as low-calorie diets (31). In a study comparing low-carbohydrate and low-fat diets over a 12-wk period, the former was shown to be more effective over the short term but with similar results by the end of the study (32). Unfortunately, the scientific literature regarding the physiologic mechanisms of non-medical "fad" diets is scarce. Current recommendations are focused on programming intake and avoidance of caloric excess (degree of recommendation: C–D), suggesting that obese children and adolescents consume a mixed and varied diet with a moderate caloric restriction (4,33). To help children understand these recommendations, programs such as the Traffic Light or Stoplight diet were developed in the 1970s by Epstein et al. (34). Nutritional intervention strategies are strongly suggested to be combined (degree of recommendation: A) (4,30).

Physical activity. Obese children and adolescents perform less physical activity than their nonobese peers. Reducing sedentary activities and increasing physical activity have positive effects on body composition and obesity-associated metabolic comorbidities (35) and are highly recommended in available clinical practice guides (degree of recommendation: A). Programming of collective family physical activities is also recommended (degree of recommendation: B) (4,30). To achieve the first objective, the removal of the television and/or computer from the child's room is recommended (degree of recommendation: A) (4,30). The time and type of sedentary activity should be limited and previously programmed with the parents. Walking to all destinations and avoiding the use of mechanical means are recommended when possible. The type of physical activity should be adapted to the child's age and be attractive and fun for him or her, changing as the child matures. In the second stage, in accordance with his/her acquisition of skills and improved physical condition, the duration and intensity of the activity can be gradually increased, with team activities included when the child feels sufficiently confident (36). There is no consensus regarding the frequency, duration, or intensity of the physical activity that should be performed.

Exceptional treatments in childhood obesity

The use of drugs, surgery, or other treatments has evolved because of the increased prevalence of severe comorbidities, including those that are potentially lethal. However, data regarding the use and outcome of these treatments in children are severely limited.

Pharmacologic treatment. The only active substance that is currently available for treatment of obesity is orlistat, an inhibitor of pancreatic and gastric lipase that interferes with the absorption of cholesterol and FFAs from the diet. The few controlled trials of this drug in adolescents reported significant variability in the reduction in BMI, with data in prepubertal children being virtually absent (37,38). There is no general recommendation regarding the use of this drug in this age group. However, a meta-analysis of available data suggested that this drug could help in conjunction with normal dietary treatment

of obese adolescents, at least during the first 12 mo (39), with the indication that lipid-soluble vitamin supplements should be used because this drug interferes with their absorption. Adverse gastrointestinal effects may also limit its use. There are no data on which to base a recommendation of an optimal dose, and the long-term effects are unknown.

Surgical treatment. There are few available data regarding bariatric surgery in children and adolescents and they are limited to the publication of case studies and expert consensus statements, with no controlled studies or data regarding the long-term results. The retrospective series analyzed by the Collaborative Group for the Study of Pediatric Bariatric Surgery is based on gastric bypass surgery, the most used technique in the United States, whereas in Europe and Australia adjustable gastric banding has been more widely used. Thus, specific recommendations regarding its use in children and adolescents cannot be made, and it should be limited to extreme cases and performed only in highly specialized centers (4,30).

For an adolescent to be candidate for bariatric surgery, expert committees require that, in addition to the anthropometric requirements (BMI >40 kg/m^2) and the presence of severe associated comorbidities, the patient's maturity, both physical (estimated by bone maturation) and cognitive, as well as his/her capacity for decision making and the family structure should be evaluated to reduce possible adverse effects both during surgery and follow-up. An additional requirement includes previous failure of intense weight-loss programs during a minimum of 6 mo (40). Hence, the percentage of children and adolescents who are candidates for bariatric surgery is very low.

Treatments under development. The fact that the administration of recombinant leptin to patients with leptin deficiency reverses their clinical picture (41) opens the door to the development of new treatments for specific conditions, such as the alteration of *MC4R* (42). However, to date, clinical trial data are limited (43,44), with modest effects on weight reduction and moderate secondary effects (flushing, nausea, vomiting, headache, insomnia, and alterations in taste) being reported. There are new drugs that have been extensively reviewed in recent publications (45–47).

Final Remarks

The incidence of obesity in children and adolescents is clearly increasing in developed countries, and our understanding of the pathophysiologic basis has advanced in recent years. Although most these cases represent common or exogenous polygenic obesity, other causes are being discovered and must be diagnosed. Indeed, we should now speak of "obesities" and not obesity. It is currently impossible to determine the precise percentage of each form of obesity because large epidemiologic studies are not yet available. Moreover, in many circumstances, the underlying cause of obesity has not been thoroughly studied. We suggest that with the increasing interest in personalized medicine, specific diagnoses of all forms of obesity will be fundamental in the near future.

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